LSRFormer: Efficient Transformer Supply Convolutional Neural Networks With Global Information for Aerial Image Segmentation

Renhe Zhang, Qian Zhang, and Guixu Zhang

***Abstract*— Both local context information and global context information are essential for the semantic segmentation of aerial images. Convolutional neural networks (CNNs) can capture local context information well but cannot model the global dependencies. Vision transformers (ViTs) are good at extracting global information but cannot retain spatial details well. In order to leverage the advantages of these two paradigms, we integrate them into one model in this study. However, global token interaction of ViT brings high computational cost, which makes it difficult to apply to large-sized aerial images. To handle this problem, we propose a novel efficient ViT block named long-short-range transformer (LSRFormer). Instead of mainstream ViTs designed as backbones, LSRFormer is a pretraining-free and plug-and-play module to be appended after CNN stages to supplement the global information. It is composed of long-range self-attention (LR-SA), short-range self-attention (SR-SA), and multiscale-convolutional feed-forward network (MSC-FFN). LR-SA establishes long-range dependencies at the junction of the windows and SR-SA diffuses the long-range information from window boundary to internal. MSC-FFN can capture multiscale information inside the ViT block. We append the LSRFormer block after each CNN stage of a pure convolutional network to build a model named ConvLSR-Net. Compared with existing models which combine CNN and ViTs, our model can learn both local and global representations at all stages of the model. In particular, ConvLSR-Net achieves state-of-the-art (SOTA) results on four challenging aerial image segmentation benchmarks, including iSAID, LoveDA, ISPRS Potsdam, and Vaihingen. The code has been released at https://github.com/stdcoutzrh/ConvLSR-Net.**

***Index Terms*— Global dependencies, remote sensing, semantic segmentation, vision transformer (ViT).**

I. Introduction

SEMANTIC segmentation is a common task in computer vision (CV) field, and the same to aerial image processing. With the rapid development of aerospace technology and electronic sensor technology, a large volume of high-resolution aerial images with rich information have been collected. The correct extraction of objects in aerial images plays a key role in many real-world applications, such as urban planning, anomaly detection, and land change detection [1], [2].

Traditional methods use manually designed features (e.g., texture and color) to segment aerial images, but the objects in aerial images usually vary greatly in shape and color distribution. Handcraft features lack semantic information, and thus, they cannot segment aerial images well. With the development of deep learning (DL) [3] technologies in recent years, convolutional neural networks (CNNs) [5], [6], [7] were first applied to general image classification task [4]. In the field of remote sensing image processing, CNN is also widely used. CCR-Net [8] applied CNN to multimodal remote sensing image classification tasks and achieved excellent results. Through the mechanism of local window sliding filtering, CNN-based segmentation models [9], [10], [11], [12], [13] can efficiently extract local spatial features and learn rich semantic information. These methods also achieve better segmentation results than traditional feature methods.

However, due to the fixed size of convolutional kernels, they encounter the problem of a local receptive field. Compared with images in general scenes, objects in aerial images have the characteristics of large scale and appearance differences [39], [40]. Thus, local context is usually not effective enough. If long-range dependencies can be established, the accuracy of aerial image segmentation can be greatly improved [14], [15], [16], [17], [18], [25], [26], [33], [39]. To capture long-range context information, a series of nonlocal (NL) methods [14], [15], [16], [17], [18], [25], [26], [33], [39] have been proposed, which have proven the effectiveness of global information for semantic segmentation tasks. These attention-based modules can establish long-range dependencies to a certain extent. However, considering the computational cost, their modules are always applied to a few deep layers of the model, where the size of feature map is very small. They still rely too much on preprocessed convolution layers and cannot effectively learn global representation at the early stages of the model.

Transformer was first proposed in the nature language processing (NLP) task. Standard vision transformer (ViT) [19] applied it to general image tasks and achieved good results. It processes images from 2-D to 1-D sequences and achieves information exchange through self-attention (SA). Thanks to this global information exchange capability, the

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ViT variants have also been applied in remote sensing image processing tasks. ExViT [53] proposed the extended ViT and achieved advanced results in multimodal remote sensing image classification task. Recently, ViT-based methods [20], [21], [22], [23], [49] also demonstrate effectiveness in global information interaction for remote sensing image segmentation. Despite the excellent performance, they also have limitations: the first is high computational cost. Mainstream ViTs [20], [21], [22] usually downsample the image first to alleviate this problem, but this operation inevitably leads to the loss of low-level information. Compared with the general image, the aerial image is usually larger, which will further make it difficult for ViTs to process. Finally, limited by the computer mechanism of SA, it is difficult for them to retain local spatial details in the process of feature extraction [48].

CNNs are good local context information extractor but cannot model the global dependencies. ViTs are good at extracting global information but cannot retain spatial details well. Thus, many works have attempted to integrate CNNs and ViTs into one model to combine the advantages of both of them. These works have proven that ViTs can supplement global information for CNN and achieved good results. However, by summarizing the existing methods of combining these two paradigms, we find that these models have a common disadvantage, that is, like the NL modules [14], [15], [16], [17], [18], [25], [26], due to the high computational cost of ViTs, their combination can only occur in a few stages of the model. In other words, their combination methods can make some stages of the model have both local and global context information, but not all stages. This leads to a lack of effective global information exchange in some stages of their model, and their model cannot fully extract global information.

The standard SA calculates the relationship between a token and all other tokens for global information exchange. This leads to a quadratic complexity in the number of tokens, making it unsuitable for high-resolution remote sensing images [20], [21], [48]. Furthermore, due to the fixed window size and token dim, previous ViT blocks [20], [21], [22], [23], [49] lack internal multiscale information. In the semantic segmentation task of remote sensing images, there is a significant difference in the size of the targets (e.g., cars and buildings). The lack of multiscale information within the ViT block will hurt the performance of semantic segmentation [22].

Inspired by these outstanding works, in this article, we propose a block-level tandem structure combination with CNN as the core and ViT as the supplement for aerial image segmentation. As shown in Fig. 1(d), different from existing methods that combine CNN and ViTs [Fig. 1(b)–(d)], we realize the combination at every stage of the model. This ensures that our model can fully capture global information. Then, we design an efficient ViT block named long-short-range transformer (LSRFormer) to ensure the efficiency of our ConvLSR-Net architecture. It captures long-distance dependencies through new token interaction method while maintaining low computational costs. Compared to previous advanced ViT blocks, our LSRFormer is more friendly to large-sized feature maps as it can be added to all stages of the model. In addition, it is plug-and-play and pretraining-free. Therefore, it can be flexibly added after each CNN stage without the need for specially designed feature fusion modules (FFMs).

The contributions of this study are the following three points.

1) We have summarized the existing methods that combine CNNs and ViTs for remote sensing image segmentation and found their common flaw, that is, they only insert ViT blocks into serval stages of the model due to the highly increased computational cost. This prevents their model from fully extracting global information. To address this issue, we build our ConvLSR-Net architecture, which combines CNN and Transformer with block-level tandem structure. Our model can learn both local and global representation at all stages of the model. Benefiting from this, it achieves state-of-the-art (SOTA) results on four widely adopted aerial image semantic segmentation datasets with different characteristics.
2) To realize the long-range information interaction more efficiently, we design a novel and efficient two-stage SA computation method. It is composed of long-range SA (LR-SA) and short-range SA (SR-SA). LR-SA establishes long-range dependencies at boundaries of adjacent windows, which can obtain approximate global representation with strip sparse sampling and lower computational cost. Then, SR-SA diffuses the long-range information from window boundary to internal. In this way, the local representation of SR-SA can be guided and optimized by global information from LS-SA. Through this design, our module can more efficiently interact with global information compared to the previous advanced SA modules. This ensures the efficiency of our ConvLSR-Net architecture.
3) To address the issue of the lack of multiscale information within the ViT block, we propose a multiscale-convolutional feed-forward network (MSC-FFN). It captures multiscale information through internal multiscale convolution. Based on LR-SA, SR-SA, and MSC-FFN, we propose an efficient and pretraining-free ViT block named LSRFormer. It can be added to all stages of the model (even in stages with large-sized feature maps) to fully capture global information and multiscale features at a lower computational cost.

II. Related Work

A. Segmentation With CNN Methods

Fully convolutional network (FCN) [9] is a milestone work in semantic segmentation task. Subsequent studies [10], [11], [12], [13], [14], [15] mainly focused on reducing the loss of detailed information during the downsampling process and expanding the receptive field. For example, DeeplabV3+ [14] introduces atrous convolution and Atrous Spatial Pyramid Pooling (ASPP) module to capture information with different receptive fields. These methods enlarge the receptive field,

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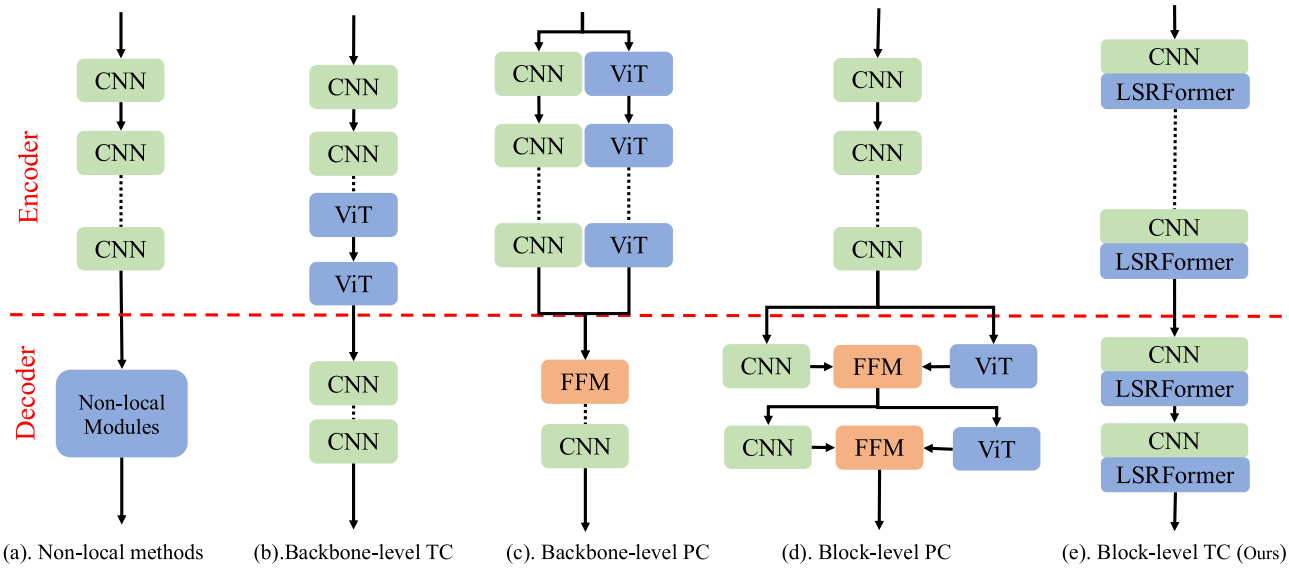


Fig. 1. Comparison with previous architectures to supplement global information for CNNs, such as (a) CCNet [18], (b) TransUNet [24], (c) ST-UNet [36], and (d) UNetFormer [34]. FFMs are specially designed FFMs. TC is a tandem combination and PC is a parallel combination. Due to computational cost considerations, their combination of the CNNs and ViTs only occurs in serval stages of the model. The architecture we proposed (e) can highly efficiently learn both local and global representation at all stages by the newly designed LSRFormer module.

but it is still hard to capture global information. Therefore, some NL methods [14], [15], [16], [17], [18], [25], [26] have been proposed. For example, HMANet [25] introduced the hybrid multiple attention to capture global information. These methods can establish global dependencies to a certain extent. However, they are usually only appended in the last several stages of the model due to the large computational cost of their modules. This prevents them from fully capturing the global information of previous stages of the model and brings limited performance improvements.

B. Segmentation With ViT Methods

As powerful ViT backbones [19], [20], [21], [22] are proposed, some ViT-based segmentation models [23], [24], [32], [37], [49] have also been proposed. For example, Segmenter [37] and SETR [49] introduced the ViT [19] as the backbone to capture the global context information. Compared with the CNN-based models, these ViT-based methods can fully capture global dependencies of image patches. However, the local spatial detailed information, meanwhile, is ignored [48].

CNN- and ViT-based methods have their own advantages but also have limitations. Therefore, many works combine them into one model. The first typical combination structure is parallel combination (PC) at the backbone level [35], [36]. As shown in Fig. 1(c), this type of model has two branches: the CNN branch extracts local information and the Transformer branch models global dependencies. The fusion module is designed to fuse the outputs of each stage of the two branches. The second structure is the tandem combination (TC) at the backbone level [24]. As shown in Fig. 1(b), this kind of model is actually dominated by CNNs, which provides a large number of convolutions in front of the network to complete the local feature extracting and then sends the feature maps to several Transformer blocks for global context modeling. Finally, the PC at the block level [34]. As shown in Fig. 1(d), this kind of model achieves the PC of CNNs and Transformer within the block of specific stages.

These models combine the ViT and CNNs to build local and global dependencies in the model. However, they can only be combined in partial stages of the model due to the high computational cost of the ViT, as with NL methods. For example, TransUNet [24] only appends ViT blocks to the last several stages of the encoder, and its decoder is still pure CNNs [see Fig. 1(b)]. UNetFormer [34] combines CNNs and Transformer at each stage of the decoder, but its encoder is still pure CNNs [see Fig. 1(d)].

This makes it inevitable for these models to lack global information interaction in some stages of the model, and in other words, these models cannot fully establish global dependencies. Unlike these previous works, our study realizes the strategy of tandem structure combination with CNNs as the core and Transformer as the supplement at each stage of the model. In addition, our module can be easily inserted after the CNN stage of the model, without the need to design a special FFM. Through this design, our method can fully explore the global information at all stages of the model.

C. Efficient Window-Based ViT

The most relevant to our method to efficiently model global dependencies are window-based ViTs. Global SA brings a quadratic computation complexity. One solution to reduce the computation cost is splitting the feature map into different regions and performing SA in local windows. The typical works are [20], [34], [45], and [46]. These methods follow the pipeline of short-range to long-range, that is, they first establish local dependencies in windows and then realize global dependency modeling through interaction between windows. For example, Swin Transformer [20] proposed Window MSA (WMSA) to perform SA within the

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local window and Shifted-WMSA (SWMSA) to complete information interaction in different windows. MT-UNet [46] first performs similarity calculation within the divided window, then aggregates the tokens in each window, and finally performs global SA on the tokens after the quantity is reduced the Transformer block. The window size and token dimension of these window-based ViTs are fixed. Thus, they lack the ability to identify targets of different scales.

Unlike existing methods which combine CNNs and ViTs, our solution is to realize the block-level tandem structure combination of these two paradigms. After effective local feature learning of CNNs, connecting with local SA within the window is redundant. To solve the above problems, we creatively proposed a method to establish context dependencies from long range to short range. Long-range dependencies can obtain approximate global representation with strip sparse sampling and lower computational cost. Then, the local representation of short range can be guided by global information from LS-SA. We also proposed a novel FFN to capture multiscale information inside the Transformer block.

III. Proposed Method

A. Overview of ConvLSR-Net

In order to more effectively learn global and local information at all stages of the model, we combine CNN and ViT at each stage. Specifically, we build a novel network with block-level tandem structure combination of CNNs (as the core) and Transformer (as the supplement) to take the advantage of both of them. The overall architecture of our proposed ConvLSR-Net is illustrated in Fig. 2. The main difference between our SA module in LSRFormer block and the previous window-based ViTs (the pipeline from short range to long range) is that it establishes context dependencies from long range to short range. The MSC-FFN is used to make up for the lack of multiscale information inside the LSRFormer block. It enhances the capability of extracting small objects in aerial images through multiscale preconvolution. In the following, we describe the novelties of our methods one by one.

B. LSRFormer Block

The global SA in the standard Transformer [19] uses three linear transform matrices W_q, W_k, and W_v to obtain query, key, and value, respectively. Generally, the tokens in query and key are individually matched by point multiplication calculation. The calculated result attention maps reflect the similarity information among tokens, and value represents the important information extracted from each token. The attention map is then multiplied by value to obtain the output. By performing such calculation on all image tokens, SA ensures a global information exchange. In other words, unlike CNNs, SA has a global receptive field. SA can be formulated as follows:

$$Q = XW_q \tag{1}$$

$$K = XW_k \tag{2}$$

$$V = XW_v \tag{3}$$

$$\mathrm{SA}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{4}$$

where d donates the dimension of each image token. However, global information interaction brings quadratic complexity with respect to the number of tokens. For efficient modeling global dependencies after the CNN layer, we proposed LSRFormer, which first establishes long-range dependencies at the junction of the windows and then diffuses the obtained long-range information from window boundary to inside.

As illustrated in Fig. 3, the LSRFormer block is composed of LR-SA, SR-SA, and MSC-FFN. The input feature map $X \in R^{C\times H\times W}$ would be convoluted to $X \in R^{(C/4)\times(H/2)\times(W/2)}$ first.

Then, the feature map is split into nonoverlapping local windows (red boxes) according to the fixed set size. Here, due to that the size of the feature map (such as 128 or 256) is usually even, we also set the window size to even. We have tried the window size as two, four, and eight. We found that setting it to four yields the best results, and the computational cost is also relatively appropriate.

As shown in Fig. 4, taking a feature map with size of 8×8 as an example. Two red lines divide the feature map into four 4×4 windows. Then, we calculate the matrix indexes located at the junction of windows (yellow box and orange box). Here, we will get indexes as $\{3, 4\}$. Based on these indexes, we extract the tokens within the two boxes and use convolution to further reduce the number tokens. A feature map with size of $H \times 2$ will be convolved into $(H/2) \times 1$. H is the height of the original feature map, here it is 8. LR-SA performs the standard SA calculation on these reduced tokens. Finally, we upsample and restore the LR-SA learned tokens to its original size (from $(H/2) \times 1$ to $H \times 2$) and add it back to the original feature map using the previously extracted indexes ($\{3, 4\}$) to provide local diffusion of global information with SR-SA.

SR-SA performs standard SA within local windows with size of 4×4 to diffuse the long-range information. This greatly reduces the computational cost required compared to global SA.

MSC-FFN captures multiscale information inside the transformer block. Mathematically, for an input feature map $X \in R^{C\times H\times W}$, the overall process of LSRFormer can be formulated as

$$X = \mathrm{Reshape}(\mathrm{Conv}(x)) \tag{5}$$

$$z_{\text{long-range}} = \mathrm{LRSA}(X) + X \tag{6}$$

$$z_{\text{short-range}} = \mathrm{SRSA}\left(z_{\text{long-range}}\right) \tag{7}$$

$$z = \mathrm{Upsample}\left(\mathrm{MSCFFN}\left(z_{\text{short-range}}\right) + z_{\text{short-range}}\right) \tag{8}$$

where z refers to the module output.

C. Multiscale Convolutional FFN

Due to the fixed window size and token dim, previous window-based ViT blocks lack internal multiscale information. Thus, in order to introduce multiscale features inside the LSRFormer block, as shown in Fig. 5, we capture this information through convolution with three kernel sizes

$$x_1 \in R^{\frac{C}{8}\times\frac{H}{2}\times\frac{W}{2}} = \mathrm{Conv}(x; k_1) \tag{9}$$

$$x_2 \in R^{\frac{C}{16}\times\frac{H}{2}\times\frac{W}{2}} = \mathrm{Conv}(x; k_3) \tag{10}$$

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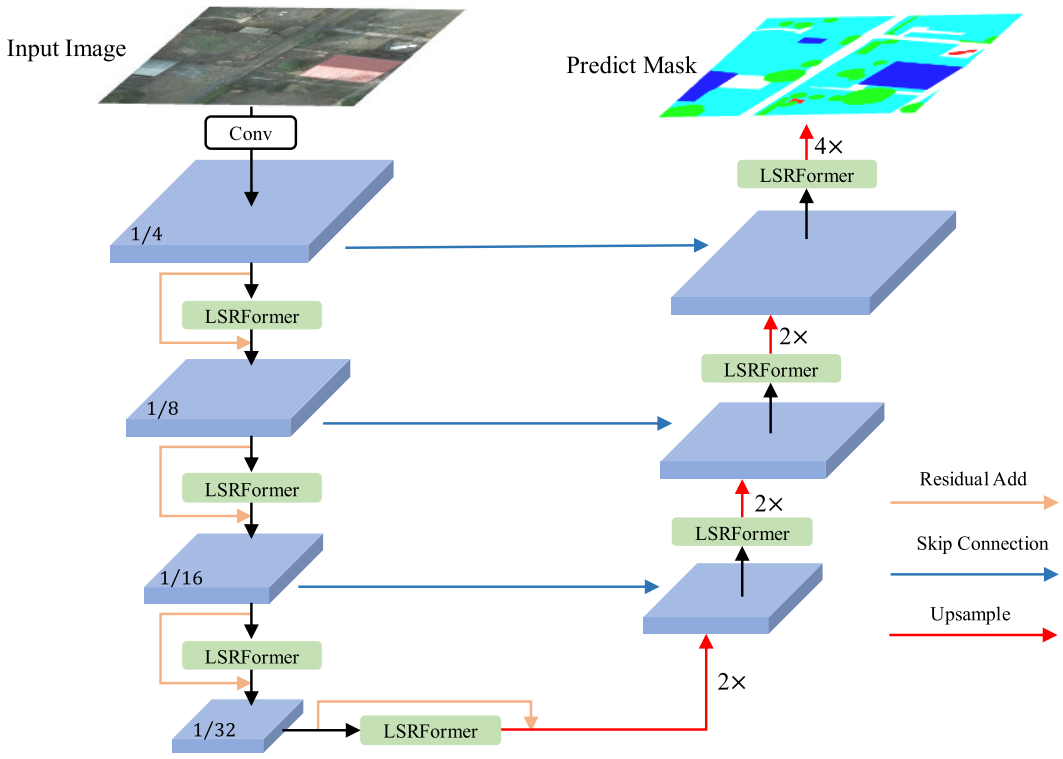


Fig. 2. Overall structure of our ConvLSR-Net. We design it by appending our LSRFormer block after all CNN stages of UNet-like framework.

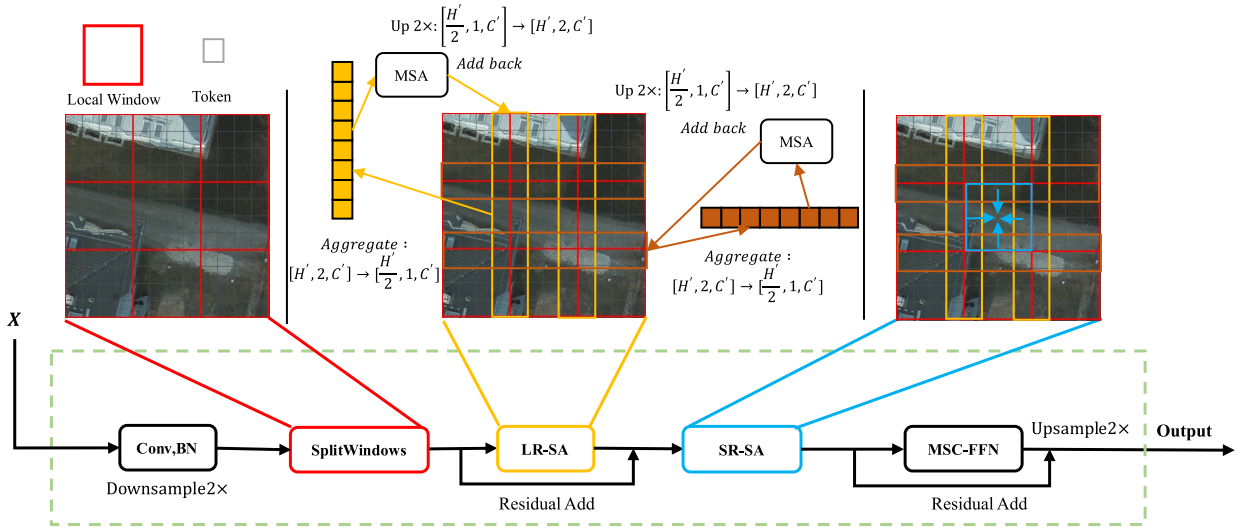


Fig. 3. Illustration of LSRFormer block. In LR-SA, yellow and orange boxes establish long-range context dependencies at the vertical and the horizontal junctions of windows, respectively. In SR-SA, blue boxes (here, we only draw one blue box for easy observation and understanding) diffuse the long-range context information in each local window from boundary to interior.

$$x_3 \in R^{\frac{C}{16}\times\frac{H}{2}\times\frac{W}{2}} = \mathrm{Conv}(x; k_7) \tag{11}$$

$$x_4 \in R^{C\times\frac{H}{2}\times\frac{W}{2}} = \mathrm{Conv}(\mathrm{Concate}(x_1, x_2, x_3); k_1) \tag{12}$$

$$x_5 \in R^{\frac{C}{4}\times\frac{H}{2}\times\frac{W}{2}} = \mathrm{Conv}(x_4; k_1) \tag{13}$$

where $\mathrm{Conv}(x; k_i)$ is the convolutional layer with kernel size $i \times i$ and x_j refers to the jth steps output of module.

IV. Datasets and Experiment Settings

A. Datasets

Following the previous work [29], we perform ablation studies on the iSAID [41] dataset to evaluate the effects of each component of our model and compare them with other methods. We benchmark more models proposed for aerial image segmentation on other four public aerial datasets: LoveDA [42], ISPRS Vaihingen [43], and Potsdam [44] to further evaluate the effectiveness of our model.

1) *iSAID:* It is the largest aerial image segmentation dataset. It includes 15 foreground classes and a background class. Its training, validation, and test sets separately involve 1411/458/937 images.
2) *LoveDA:* It is a large aerial image segmentation dataset in which the images consist of rural and urban areas.

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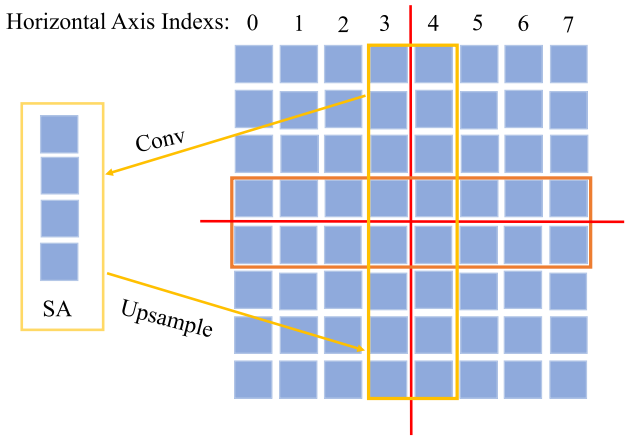


Fig. 4. Illustration of LR-SA.

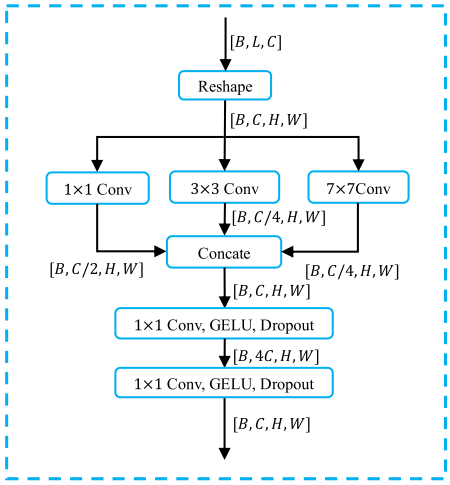


Fig. 5. Illustration of MSC-FFN.

This brings considerable challenges such as complex objects and inconsistent class distributions. Specifically, the dataset contains 5987 fine-resolution optical aerial images with a size of 2048 × 2048, including seven land cover categories. The ground sampling distance (GSD) is 0.3 m.

3) *ISPRS Vaihingen and Potsdam:* These two datasets are the most used by researchers in aerial image segmentation task. The Vaihingen dataset consists of 33 fine spatial resolution TOP image tiles (GSD 9 cm) at an average size of 2494 × 2064 pixels. The dataset includes five foreground classes and one background class. The Potsdam dataset contains 38 very fine spatial resolution TOP image tiles (GSD 5 cm) with a size of 6000 × 6000 pixels. It involves the same category information as the Vaihingen dataset.

B. Experimental Details

1) Dataset Partitioning: For a fair comparison, we follow the previous works [29], [34] to process corresponding datasets. Specifically, for the iSAID dataset, we cropped images of 896 × 896 based on overlapping pixels of 384. It generates 33 978 images for training and 11 644 images for validation. For the ISPRS Potsdam and Vaihingen datasets, we crop images to 1024 × 1024. We use the widely adopted 16 images for training and 17 for testing in the Vaihingen dataset, while the setting in the Potsdam dataset is 24 tiles for training and 14 tiles for testing. For the LoveDA dataset, since the original images are 1024 × 1024, we did not crop them.

TABLE I
COMPARISON WITH THE SOTA METHODS ON THE iSAID DATASET

Method	Backbone	mIoU
DenseASPP [15]	ResNet50	57.3
PSPNet [11]	ResNet50	60.3
DeeplabV3+ [12]	ResNet50	61.2
SemanticFPN [14]	ResNet50	62.1
FarSeg [26]	ResNet50	63.7
HMANet [25]	ResNet50	63.9
SFNet [13]	ResNet50	64.3
VB+R-UperNet [27]	ViTAE-B	64.5
SwinUperNet [20]	Swin-T	64.6
ViTAEv2-S-UperNet [28]	ViTAEv2-S	65.3
PFNet [29]	ResNet50	66.9
SegNeXt-L [30]	MSCAN-L	70.3
ConvLSR-Net (Ours)	ResNet50	68.9±0.24
ConvLSR-Net (Ours)	ConvNeXt-S	**70.8±0.11**

2) Experimental Settings: Following the previous work [34], we adopt the AdamW for the optimizer with cosine learning rate change strategy. The base learning rate is $6e^{-4}$. We trained our model on two RTX 3090 GPUs and Ubuntu 20.04 operating system. For the Potsdam and Vaihingen datasets, the training epochs were set as 105. For the iSAID and LoveDA datasets, the training epochs are 60 and 30, respectively. We adopt some common data augmentation, including random flipping, random scaling, and random cropping. We used the pretrained weights in ImageNet [4] to initialize the CNN backbone, while LSRFormer blocks and the decoder are randomly initialized.

3) Other Models for Comparison: We selected a comprehensive set of benchmark methods for quantitative comparison, including SOTA general [8], [9], [10], [11], [12], [15], [17], [20], [21], [27], [34], [35] and aerial [22], [23], [24], [25], [26], [29], [31], [32], [33], [37] image segmentation methods. We adopt a pure convolutional UNet as the baseline model, which only models the local contextual information at all stages of the model.

C. Evaluation Metrics

Following the previous works, for the iSAID dataset and LoveDA dataset, we adopt mean intersection over union (mIoU) as indicators. For the Vaihingen dataset, we adopt mIoU, overall accuracy (OA), and mean $F1$ score (mF1) as indicators. The OA, mF1, and mIoU are defined as

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FP}+\mathrm{TN}+\mathrm{FN}} \tag{14}$$

$$\mathrm{mF1} = \frac{1}{k+1}\sum_{i=0}^{k}\frac{2\mathrm{TP}}{2\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \tag{15}$$

$$\mathrm{mIoU} = \frac{1}{k+1}\sum_{i=0}^{k}\frac{\mathrm{TP}}{\mathrm{FN}+\mathrm{FP}+\mathrm{TP}} \tag{16}$$

where k is the number of target segmentation categories, TP is the true positive, TN is the true negative, FP is the false

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TABLE II
COMPARISON WITH THE SOTA METHODS ON THE POTSDAM DATASET

Method	Imp.surf	Building	Low.Veg	Tree	Car	mF1	OA	mIoU
SR-A-FCN [40]	91.33	94.70	86.81	83.47	94.52	90.17	88.59	82.38
DeeplabV3+ [12]	92.95	95.88	87.62	88.15	96.02	92.12	90.88	84.32
DDCM-Net [51]	92.90	96.90	87.70	89.40	94.90	92.30	90.80	-
PSPNet [11]	93.36	96.97	87.75	88.50	95.42	92.40	91.08	84.88
BoTNet [50]	93.13	96.37	87.31	88.01	95.79	92.12	90.76	85.62
CCNet [18]	93.58	96.77	86.87	88.59	96.24	92.41	91.47	85.65
MANet [39]	93.40	96.96	88.32	89.36	96.48	92.90	91.32	86.95
HMANet [25]	93.85	**97.56**	88.65	89.12	96.84	93.20	**92.21**	87.28
STransFuse [35]	89.75	93.92	82.91	83.61	88.51	82.08	86.71	71.46
ST-UNet [36]	79.19	86.63	67.89	66.37	79.77	86.13	-	75.97
TransUNet [24]	92.60	96.00	86.80	87.90	95.30	91.70	90.10	84.90
DCSwin [32]	94.19	97.57	88.57	89.62	96.31	93.25	92.00	87.56
Segformer [38]	93.70	97.10	88.30	89.80	96.50	93.10	91.60	87.30
FT-UNetFormer [34]	93.90	97.20	**88.80**	89.80	96.60	93.30	92.00	87.50
ConvLSR-Net (Ours)	**94.08±0.01**	97.33±0.07	88.63±0.08	**89.82±0.07**	**97.05±0.10**	**93.39±0.04**	91.92±0.04	**87.80±0.08**

TABLE III
COMPARISON WITH THE SOTA METHODS ON THE VAIHINGEN DATASET

Method	Imp.surf	Building	Low.Veg	Tree	Car	mF1	OA	mIoU
SR-A-FCN [40]	91.47	94.97	80.63	88.57	87.05	88.54	89.23	79.76
DeeplabV3+ [12]	92.38	95.17	84.29	89.52	86.47	89.57	90.56	81.57
BoTNet [50]	92.24	95.28	83.88	89.99	85.47	89.37	90.51	81.15
DDCM-Net [51]	92.70	95.30	83.30	89.40	88.30	89.80	90.40	-
PSPNet [11]	92.79	95.46	84.51	89.94	88.61	90.26	90.85	82.58
MANet [39]	93.02	95.47	84.64	89.98	88.95	90.41	90.96	82.71
CCNet [18]	93.29	95.53	85.06	90.34	88.70	90.58	91.11	82.76
HMANet [25]	93.50	95.86	85.41	90.40	89.63	90.96	91.44	83.49
STransFuse [35]	88.25	91.46	79.04	85.51	77.14	78.67	86.07	66.66
ST-UNet [36]	76.36	82.98	57.79	72.53	61.48	82.15	-	70.23
2D-Segformer [47]	90.96	94.50	81.44	87.20	81.29	87.08	88.85	77.49
SwinUNet [23]	91.50	93.00	82.50	89.00	82.90	87.80	89.00	78.40
TransUNet [24]	92.80	95.40	84.40	89.90	85.30	89.50	90.70	81.30
SwinUperNet [20]	92.80	95.60	85.10	90.60	85.10	89.80	91.00	81.80
Segformer [38]	93.00	95.70	85.00	90.00	88.90	90.50	91.10	82.90
DCSwin [32]	93.60	96.18	85.75	90.36	87.64	90.71	91.63	83.22
FT-UNetFormer [34]	93.50	96.00	85.60	**90.80**	90.40	91.30	91.60	84.10
ConvLSR-Net (Ours)	**93.84±0.04**	**96.41±0.07**	**85.82±0.06**	90.72±0.08	**90.79±0.10**	**91.52±0.03**	**91.90±0.04**	**84.56±0.06**

negative, and FN is the false negative pixel numbers of the result.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Comparison With SOTA Models

To verify the effectiveness of our model, we compare it with recently SOTAs methods on four widely used open-access datasets for aerial semantic segmentation.

1) Results on iSAID Dataset: The iSAID dataset is the largest aerial image segmentation benchmark. The challenges it brings are multiscale problems and uneven number of foreground backgrounds. As shown in Table I, our ConvLSR-Net achieves a 70.8% mIoU. It achieves the SOTA result among all previous works. It yields 0.5% mIoU improvement compared to SegNeXt [30] with powerful MSCAN-L backbone. With the same ResNet50 [6] backbone, our model exceeds the previous SOTA CNN-based model PFNet [29] by 2.0% in mIoU.

2) Results on Vaihingen and Potsdam Dataset: ISPRS Vaihingen and Potsdam are the two most widely used datasets in aerial segmentation. Many specially designed models have obtained high numerical scores on these two datasets. As shown in Tables II and III, ConvLSR-Net yields 0.30% and 0.46% mIoU improvement on two datasets respectively compared to the previous SOTA model FT-UNetFormer [34], with powerful Swin-Base [20] backbone. We present the comparison of computation cost and params quantity in Table IV. It is worth noting that our model uses only about 56% FLOPs and 71% Params of FT-UNetFormer [34]. Our model has significant advantages in terms of computational cost and actual inference speed compared to existing SOTA methods. This is also very valuable for the practical application of the model.

3) Results on LoveDA Dataset: LoveDA is a large-scale multiple domain (rural and urban) segmentation dataset. As shown in Table V, our ConvLSR-Net can not only exceed all the CNN-based models [11], [12], [14], [26], [31], [33] by a large margin in mIoU but also outperforms recently SOTA model [32], [34] proposed for aerial image by 2.37%. We extend more recently advanced segmentation methods on

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TABLE IV
COMPARISON WITH THE SOTA METHODS ON THE LOVEDA TEST SET

Method	Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU
PSPNet [11]	44.40	52.13	53.52	76.50	9.73	44.07	57.85	48.31
DeeplabV3+ [12]	42.97	50.88	52.02	74.36	10.40	44.21	58.53	47.62
SemanticFPN [14]	42.93	51.53	53.43	74.67	11.21	44.62	58.68	48.15
FarSeg [26]	43.09	51.48	53.85	76.61	9.78	43.33	58.90	48.15
FactSeg [31]	42.60	53.63	52.79	76.94	16.20	42.92	57.50	48.94
BANet [33]	43.70	51.50	51.10	76.90	16.60	44.90	62.50	49.60
TransUNet [24]	43.00	56.10	53.70	78.00	9.30	44.9	56.90	48.90
Segmenter [37]	38.00	50.70	48.70	77.40	13.30	43.50	58.20	47.10
SwinUperNet [20]	43.30	54.30	54.30	78.70	14.90	45.30	59.60	50.00
DCSwin [32]	41.30	54.50	56.20	78.10	14.50	47.20	62.40	50.60
UNetFormer [34]	44.70	58.80	54.90	79.60	20.10	46.00	62.40	52.40
Segformer [38]	43.40	62.20	56.10	81.20	19.20	**48.90**	59.20	52.90
SegNeXt-L [30]	46.04	61.40	53.97	81.56	19.90	48.58	**65.97**	53.91
ConvLSR-Net	**47.09±0.03**	**62.67±0.28**	**57.10±0.82**	**82.05±0.19**	**21.30±1.00**	48.52±0.42	64.69±0.41	**54.77±0.08**

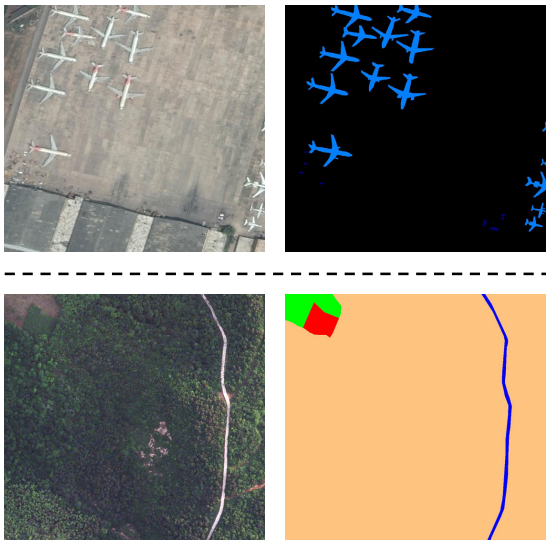


Fig. 6. Comparison of different datasets for remote sensing image semantic segmentation. (Top) iSAID dataset. (Below) LoveDA dataset.

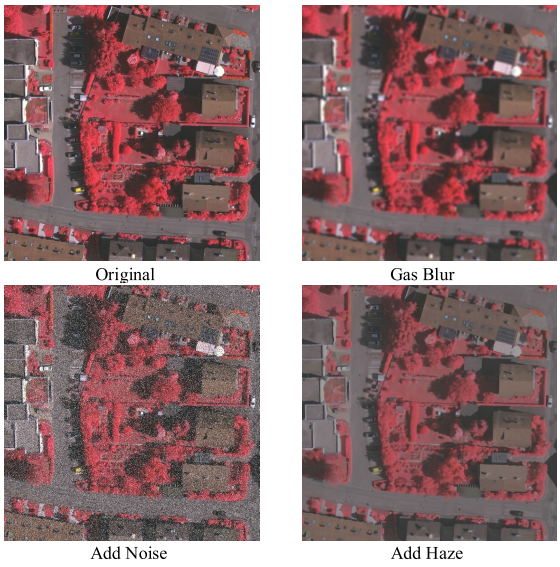


Fig. 7. Visualization of renderings after image quality degradation.

this dataset. ConvLSR-Net outperforms SegNeXt [30] and Segformer [38] by 0.86% and 1.87% in mIoU, respectively.

4) Applicability for Different Types of Datasets: There are multiple datasets in the semantic segmentation task of remote sensing images, which have different characteristics and are used for different remote sensing scientific problems. For example, as shown in Fig. 6, the iSAID dataset (top) is designed for key objects such as mobile ships, cars, and airplanes. These objects cannot reflect the characteristics of the geographical environment. However, LoveDA (below) studies fixed land cover objects that reflect biophysical substances on the Earth's surface. There are also differences in data volume, for example, although Potsdam and Vaihigen have similar categories, its data volume is much higher than Vaihigen. We have achieved SOTA results on four datasets, while SOTA segmentation methods perform well on one dataset but have average performance on another dataset. For example, on the iSAD dataset, SegNeXt [30] achieved 70.3% mIoU, but on the LoveDA, it was 0.86% lower than our model (54.77% versus 53.91%). DCSwin [32] achieved 87.56% mIoU results on the Potsdam, but it was 1.34% lower than ours on the Vaihingen dataset (84.56% versus 83.22%) and 4.17% lower than ours on the LoveDA dataset (54.77% versus 50.60%). This indicates that our method is more applicable to different types of datasets.

5) Effect of Data Degradation: It is well known that remote sensing data usually tend to suffer from various degradation and noise effects [54]. To verify the effectiveness of our model when encountering image quality degradation, we conducted experiments on the Vaihingen dataset. Specifically, as shown in Fig. 7, we added degradation strategies such as Gaussian blur (GB), noise, and add haze to the original image. As shown in Table VI, compared to the recent SOTA segmentation methods, our model exhibits better robustness. Especially on noisy images with the most severe degradation in image quality, our method outperforms other methods by 10%–20% in terms of mIoU.

B. Ablation Experiment on Modules

We first evaluate the effectiveness of our proposed methods on the iSAID [41] dataset. The results are presented in

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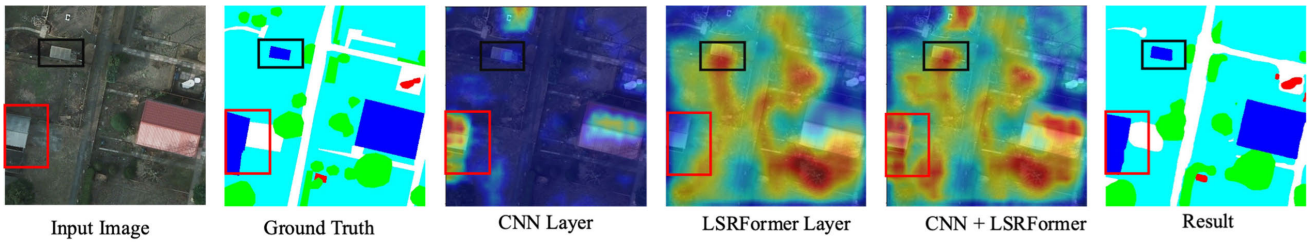


Fig. 8. GradCAM [52] at the third stage of ConvLSR-Net for the "Building" class. CNN can only focus on a small area, while our LSRFormer can establish long-range dependencies to supplement CNN with global information. When the two are combined, they can make up for the information ignored (black and red boxes) by each other.

TABLE V
COMPARISON OF PARAMS, FLOPS, AND FPS

Method	Params	FLOPs	FPS
DANet [17]	66.4M	287.5G	33
DeeplabV3+ [12]	**59.3M**	260.6G	32
ST-UNet [36]	161.0M	-	7
SwinUNet [23]	84.0M	97.7G	29
TransUNet [24]	105.9M	168.9G	27
Segformer [38]	84.6M	110.2G	20
FT-UNetFormer [34]	96.0M	128.4G	37
ConvLSR-Net (ours)	68.1M	**71.1G**	**46**

TABLE VI
ABLATION STUDY OF THE EFFECT OF DATA DEGRADATION

Method	mIoU	+GB	+Noise	+Haze
TransUNet [21]	81.50±0.20	54.12±0.18	20.31±0.43	72.42±0.69
Segformer [31]	83.01±0.13	62.13±0.27	33.28±0.57	76.95±0.65
DCSwin [29]	83.55±0.04	55.27±0.24	21.98±0.21	73.89±0.36
FT-UNetformer [35]	83.82±0.03	57.34±0.34	25.40±0.37	75.81±0.73
ConvLSR-Net (ours)	**84.56±0.06**	**63.85±0.21**	**43.04±0.52**	**78.35±0.48**

TABLE VII
ABLATION STUDY OF EACH COMPONENT OF LSRFORMER BLOCK

+LR-SA	+SR-SA	+MSC-FFN	mIoU
-	-	-	68.5±0.13
✓	✓	-	69.4±0.37
-	-	✓	69.2±0.29
-	✓	✓	69.5±0.15
✓	-	✓	69.8±0.31
✓	✓	✓	**70.8±0.11**

TABLE VIII
ABLATION STUDY OF EFFICIENCY OF LSRFORMER

Method	Blocks	Δ FLOPs	IoU
Baseline	0	-	68.5±0.13
CCA [18]	7	5.1G	69.5±0.32
DA [17]	7	23.2G	69.7±0.27
Standard Transformer [19]	6	77.1G	69.1±0.33
Swin Transformer [20]	4	67.5G	68.8±0.19
LSRFormer (ours)	**7**	**2.8G**	**70.8±0.11**

TABLE IX
ABLATION STUDY OF EFFECT OF APPENDING POSITION

Encoder	Decoder	Params	FLOPs	mIoU
-	-	58.4M	68.8G	68.5±0.13
✓	-	66.2M	70.1G	69.6±0.19
-	✓	60.4M	69.8G	69.9±0.26
✓	✓	68.1M	71.1G	**70.8±0.11**

TABLE X
ABLATION STUDY OF DIFFERENT FFNS

Method	Params	FLOPs	IoU
FFN [19]	64.4M	70.1G	69.6±0.37
Conv-FFN [21]	64.5M	70.1G	69.9±0.22
DS-FFN [22]	64.5M	70.1G	70.1 ±0.15
MSC-FFN (ours)	68.1M	71.1G	**70.8±0.11**

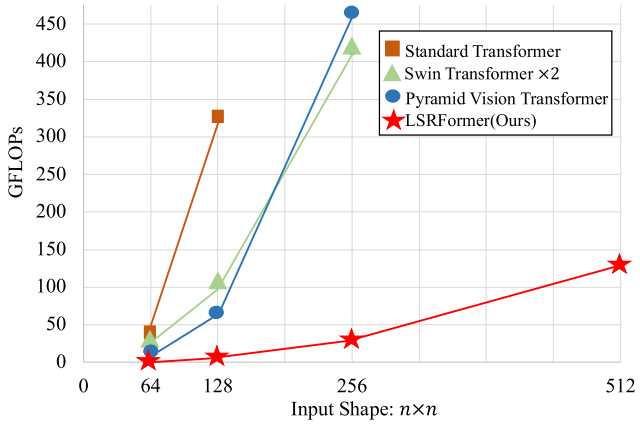


Fig. 9. Comparison between LSRFormer and recently advanced ViT blocks of numerical complexity of different input feature map shapes. Our method is more friendly to the feature map with larger size in terms of calculation cost.

Tables VII–X. Finally, we tested our module's plug-and-play performance against other CNN architectures on the LoveDA validation set in Table XI.

1) Effectiveness of the LSRFormer Block: We evaluate the performance of each component of the proposed LSRFormer block on the iSAID dataset. As listed in Table VII, only appending SA modules (second row) can improve the baseline model to a certain extent, while only appending MSC-FFN (third row) results in minimal improvement. However, the results in fourth to sixth rows show that SA + FFN is the best choice. The performance is further improved by appending the MSC-FFN and the SA modules we have proposed to model global dependencies. In Fig. 8, through retrospective visualization of the impact of the thermodynamic diagram on the model output results, our LSRFormer can effectively establish long-range dependencies and provide CNN with a supplement to global information.

2) Efficiency of the LSRFormer Block: To demonstrate the advantages of our efficient LSRFormer block, we replaced

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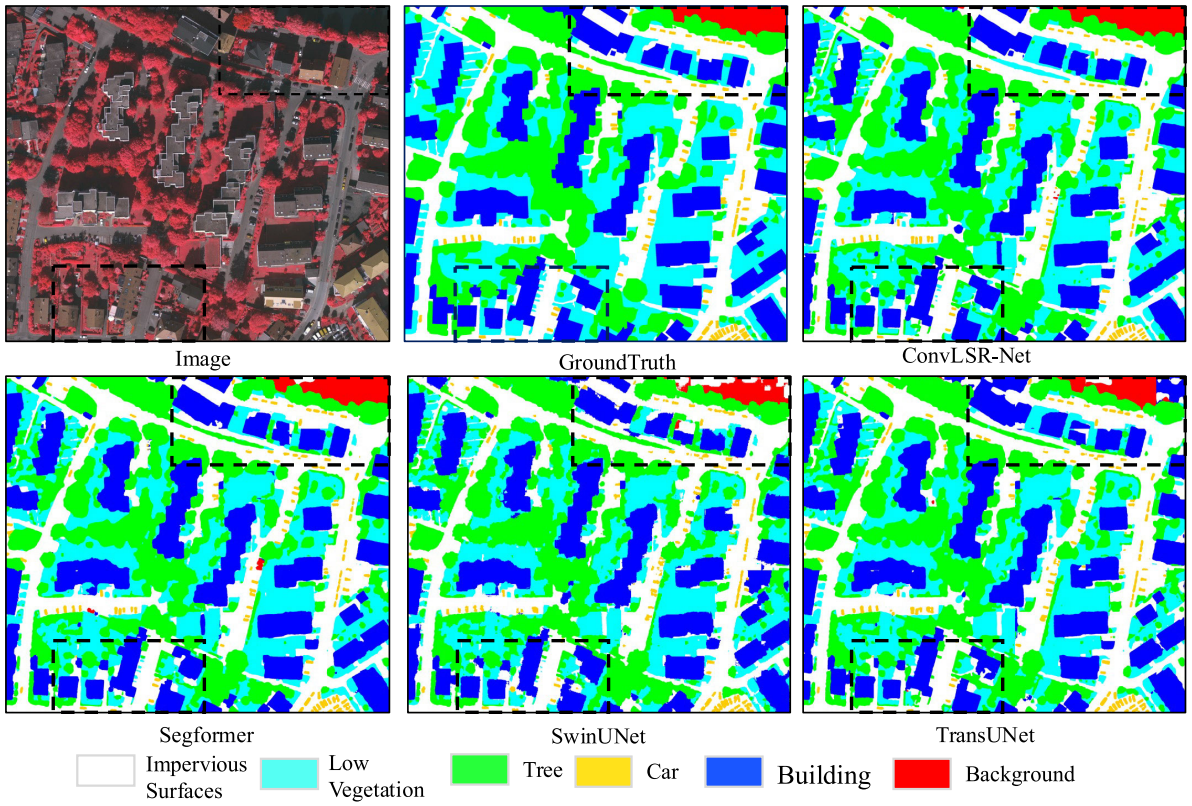


Fig. 10. Qualitative comparisons between ours and other models on the Vaihingen dataset.

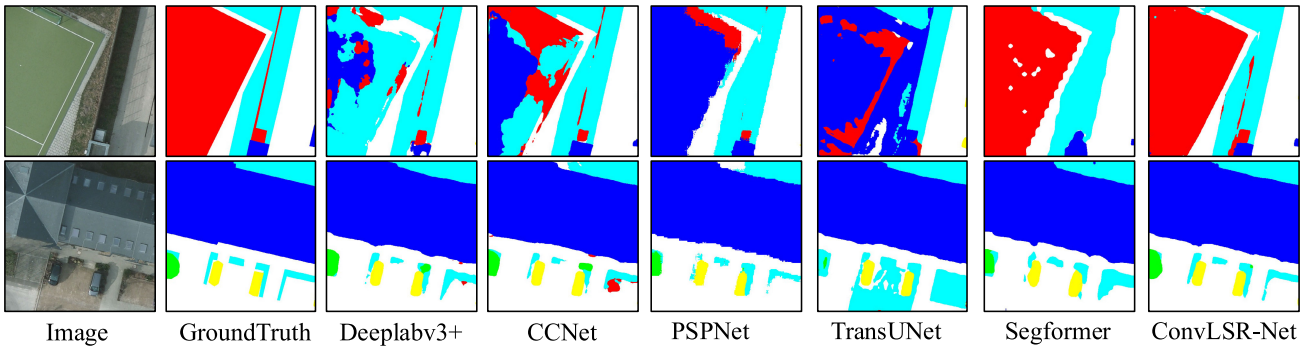


Fig. 11. Qualitative comparisons between ours and other models on the Potsdam dataset.

it with other attention modules [14], [15], [16], [17] in ConvLSR-Net for ablation studies. We have appended seven LSRFormer blocks to the baseline model, but the same appending method is not applicable to others due to the limitation of GPU memory. Therefore, for DA [14], Transformer [16], and Swin Transformer [17], we append them as much as possible within the allowable range of GPU memory. As listed in Table VIII, our method achieves the best results in terms of accuracy and additional computational cost. In Fig. 9, we further compare our LSRFormer with other advanced ViT blocks [16], [17], [18] in terms of the computation cost. As the size of input feature map increases, the FLOPs of other ViTs gradually increase exponentially, whereas the counterpart of our LSRFormer module is almost linearly increasing.

3) Effect of Appending Position: In Table IX, we explore the effect on insertion position of LSRFormer block. From the second and third rows, LSRFormer brings small improvement while only appending to the encoder or decoder. After appending it to all stages of the baseline model, we achieve the best result in the last row, which shows the necessity of supplementing global information at all stages of the model.

4) Comparison With Other FFNs: The operation in FFN [19] of the standard Transformer block [19] is all point-wise and no cross-token information interaction. In addition, our LSRFormer block lacks internal multiscale information. To complement the cross-token and local information, convolutional FFN (Conv-FFN) [21] and detail-specific FFN (DS-FFN) [22] appending a depth-wise convolution layer in the FFN. They can complete the interaction between adjacent tokens, but the FFN still lacks multiscale information internally, so as to the Transformer block. Therefore, before the linear transformation of channel dimension quadrupling, we append convolution of the three scales to build multiscale

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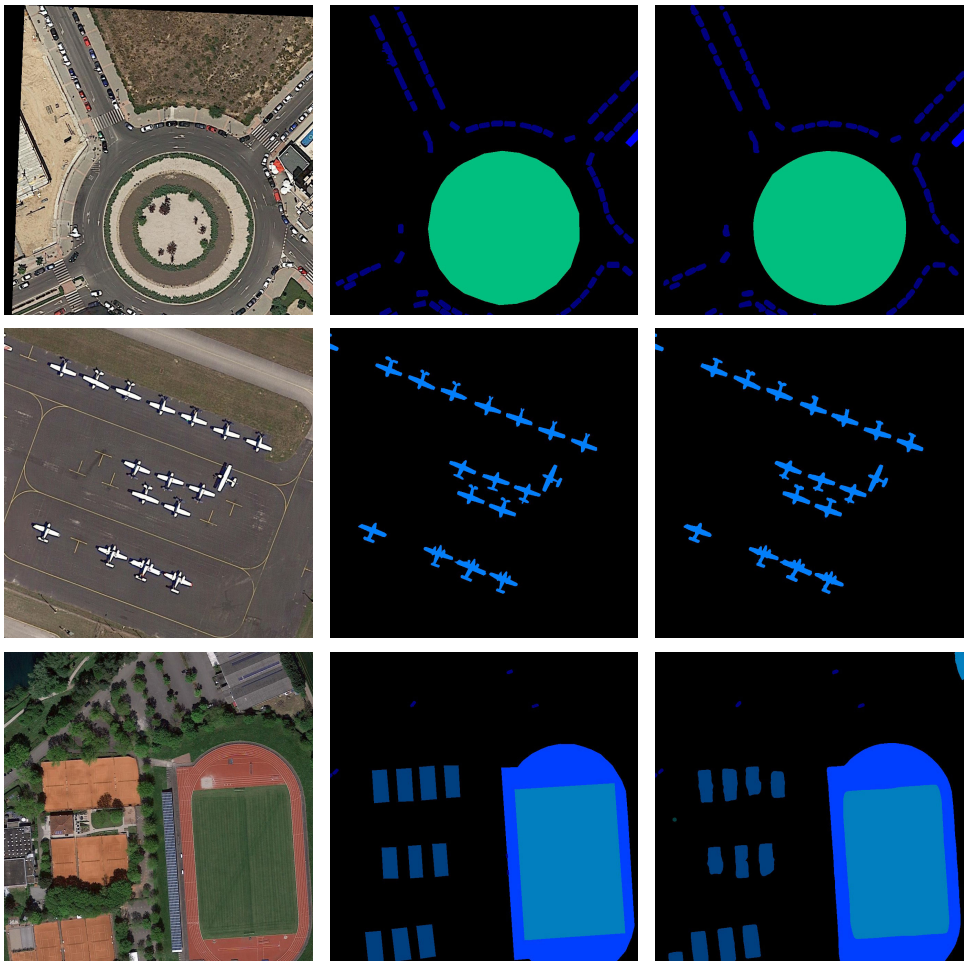


Fig. 12. Predicted results of the ConvLSR-Net on the iSAID dataset.

TABLE XI
ABLATION STUDY OF PLUG-AND-PLAY PERFORMANCE OF LSRFORMER

Method	FLOPs	ΔFLOPs	mIoU
R50-FCN [9]	33.8G	0	47.52±0.29
+LSRFormer	38.3G	4.5G	**49.59±0.33**
R50-DeeplabV3+ [12]	69.6G	0	48.86±0.41
+LSRFormer	74.5G	4.9G	**50.84±0.37**

TABLE XII
ABLATION STUDY OF DIFFERENT WINDOW SIZES

WindowSize	FLOPs	Params	mIoU
2	71.0G	68.1M	69.8±0.21
4	71.1G	68.1M	**70.8±0.11**
8	71.3G	68.1M	70.6±0.17

information inside FFN. As listed in Table X, compared with other FFNs, our MSC-FFN increases a small amount of params and FLOPs, which also brings better performance improvement.

5) Plug-and-Play Performance on Other CNN Architectures: To validate our module's plug-and-play performance on other CNN architectures and datasets, we append it to two other common segmentation architectures [9], [12] and conduct experiments on the LoveDA validation set. As shown in Table XI, our LSRFormer blocks can bring greater performance improvement by slightly increasing the computation cost.

6) Impact of Different Window Sizes: As shown in Table XII, the window size is two, the computational cost is the lowest, but due to the smaller window size, long-distance information is not effectively diffused, and the accuracy is also significantly reduced. When the window size is eight, the accuracy result is very close to four. However, it also brings additional computational costs.

C. Visualization of Prediction Results

To further verify the effectiveness of our method, in Fig. 10, we show the prediction results' comparison between our model and recently SOTA ViT-based methods [23], [24], [38] on the Vaihingen dataset. In the first row of Fig. 11, our model has better global dependencies, so as to avoid misjudgment in large areas. In the second row, benefiting from the combination ViT with CNN, our ConvLSR-Net has better detail processing ability than the previous methods. In Fig. 12, we present more prediction results on the iSAID dataset. Our model has also

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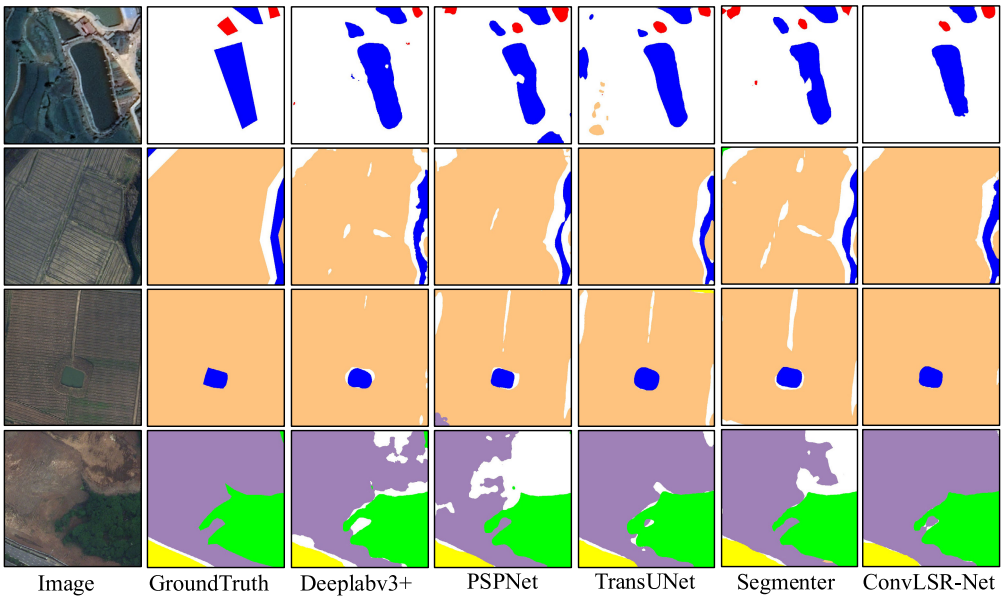


Fig. 13. Qualitative comparisons between ours and other models on the LoveDA validation set.

obtained accurate prediction results. In Fig. 13, due to the lack of test set labels, we presented the results on the LoveDA validation set. Our model achieved better results.

VI. Conclusion

Since both local context and global context information are essential to the semantic segmentation of aerial images. In this article, we combine the CNNs and the ViTs in one model and propose ConvLSR-Net architecture. Different from the previous methods, we realize a block-level TC with CNNs as the core and Transformer as the supplement at each stage of the model. We design a novel and efficient LSRFormer to ensure the efficiency of our ConvLSR-Net architecture, which first models long-range dependencies at the boundaries of adjacent windows (LR-SA) and then diffuses long-range information within the local window by SR-SA. Extensive experimental results show that LSRFormer can efficiently establish global information interaction and supplement global information for CNNs. Benefiting from this combination paradigm, ConvLSR-Net can capture both local context and global context information at each stage. This ensures that our model can fully explore global information. In addition, MSC-FFN is proposed to compensate for the lack of multiscale information inside the LSRFormer block. Our model has achieved SOTA results on four open aerial image segmentation datasets while maintaining superiority in model complexity.

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